A Leveraged Trading Strategy with Fibonacci Approach: Empirical Evidence for DJI 30 and ISE 100

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The Abstract

Considering the leverage effect in the stocks investments, this study focuses on a trading strategy by using the Fibonacci approach. For this purpose, a technical trading model based on Fibonacci approach has been developed by integrating Dow's view to the stock markets. The strategy is applied to DJI30 and ISE100 between the years of 2000 and 2005. According to the results, the return on equity performance of the above strategy with leverage ratio of 10 is much more than the market return on both DJI30 and ISE100, which is an investor's initial purpose.

Keywords: Trading strategy, stock markets, Fibonacci, leverage effect.
Jel Classification: G14, G13, G12

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1 Introduction

The random walk theory, initially mentioned by Bachelier (1900), criticizes that the future values of the stock prices are predictable. Bachelier's theory had been unnoticed for many years until Osborne (1959) derived his work independently. Some assumptions in the Bachelier – Osborne approach were developed in time and with many researches, random walk theory has become more generalized. This theory basically asserts that;

(1) successive price changes in an individual security are independent, and
(2) The price changes conform to some probability distribution (Fama, 1965).

According to the random walk theory; in the long term the mathematical expectation of the speculator is zero. There is no need to make any technical or fundamental analysis for the long term excess returns. The best strategy can only be the simple buy hold strategy.

On the other hand, in fact making higher returns than the market return is not the investors' concern at all. The main concern of an investor is; his return on equity performance. In a leveraged trading a good performing trader's equity return in a period of time can be much more than the markets return in the same period.

In this paper, it is claimed that a systematic trading strategy based on a trend following can perform much better than the market in a leveraged environment. The leverage advantage can not be used in the simple buy and hold strategy because of the absence of a stop loss point which is forming a serious risk of losing all of the related investment. The spot data of DJI30 and ISE100 has been observed between 1988 and 2005. A technical price behavior has been developed for those markets. A trading strategy which is consistent with the defined behavior of DJI30 and ISE100, has been developed and applied between 2000-2005, and it is concerned with the return on equity of the investor in a leverage trade, rather than making higher returns than the market return.

In fact the explained price behavior is a part of the trading strategy developed with observation rather than a statistical price-trend model with a satisfactory methodology. Taylor (1980), defined formal models and a stochastic process for price trends, rather than a trading rule analysis as this paper does. He defined and tested a new statistical model applicable to the

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London commodity markets and one international currency market, which is including a price-trend term as an alternative to the random walk model.

A statistical evidence for trends is not this paper's concern since the term of trend is just a tool for generating a good performing trading strategy for stock indexes especially DJI30 and ISE100. Here, the trends in the stock markets are used efficiently in order to increase the return on equity, and then the behaviors of those trends are explained by using our Fibonacci approach.

2 Non-Leveraged versus Leveraged Trading

Using of leverage in a trading means that the rules of the game have been changing. In that case, the perception of return and risk of speculators is not the same as with non leveraged trading.

In a non-leveraged trading, the return on equity is significantly correlated to the market return. In the long term, the traders' return is the some percentage of the market return. Some successful traders may have more return than the market return but generally when dividends and transaction costs are taken into account, the simple buy hold strategy is much more profitable than the other strategies which generate many transactions.

Random Walk Theory claims that, in a random walk market in the long run, technical analysis is useless and can not provide abnormal returns greater than those of simple buy and hold strategy. Fama (1965) used thirty stocks of Dow Jones industrial average between the years of 1957 and 1962 for the test of random walk. He also compared some mechanical trading strategies with simple buy and hold strategy. His results were consistent with the claims of the Random Walk Theory. The mechanical trading systems which are just searching a specific price action in order to generate buy and sell signals without understanding the big picture and market behavior, can not make greater returns than the market return in a random walk data.

The most important factor for a non-leveraged trading strategy can be summarized by taking s much as possible return from the market return.

In the case of leveraged trading, the return on equity may be 10 to 100 times of the market return on a single trade according to the leverage ratio. Generally the leverage possibility can be used on the futures markets, rather than the spot markets. In the futures markets the trader should not be concerned about getting a satisfactory return. If the trader is right in his bets the futures

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markets give returns above a fair return, with the help of the leverage availability. On the other hand a trader can never be right always on his bets. There will always be some losing trades beyond the winning ones. Trading in a leveraged environment can be called as playing a game with a two sided sharp sword. One side is expected return, the other side is risk, and both of them can be in great amounts. The return side makes these markets attractive to the traders but in fact the risk side can not be tolerated by a human being, if the suitable risk control methods are not implemented. This means that, a trader should limit his expected return in order to limit the risk as well.

The risk control method should be in harmony with the trading strategy and the investor's personality. Accountability is very important for a trading strategy, because if the strategy is analyzed with past data, the weaknesses and strengths can be understood better. The optimum risk and expected return level can be arranged according to these weaknesses and strengths.

A risk control method should avoid a trader to lose his discipline in the decision making even if the worst case scenario is seen in the market. The trader should never be threatened by the market to lose all of his wealth or equity.

3 The Price Behaviour of DJI30 and ISE100

By analyzing the data of DJI30 and ISE100 from 1988 to 2005 a price behavior is developed for these markets, which is explaining the behavior of the trends in the stock market.

The main point for this behavior comes from the fact that, the equity markets move in a growing rhythm. This growth is the sum of all trends generated by people. By analyzing these trends, following model is developed.

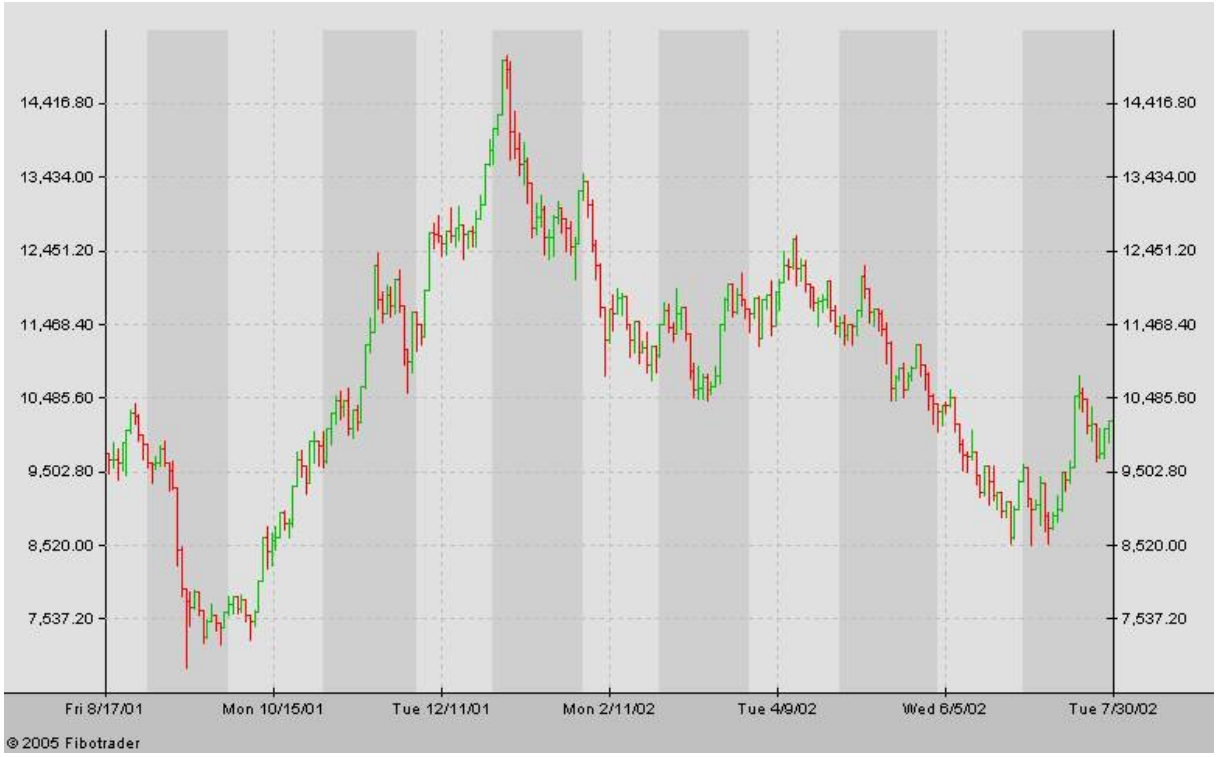
The great bull markets which stand for years are composed of some strong trends that last for months. There are also some pauses between those trends. The model calls those trends as medium term trends, the most important property of those moves are not only their size in price but also the size in time. In other words, the sudden great price changes does not form a medium term trend, because those prices after sudden movements can not survive and decrease suddenly as they increase. The medium term trends are consistent with Dow's definition of uptrend. Dow defined an uptrend as a situation in which, there is a pattern of rising peaks and troughs (Murphy, 1999). This definition also confirms the need of size in time, for the medium term trends.

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The growth model's way of looking to the market does not include the term downtrend, because any move to the downside can not be independent from the previous up moves. Those moves which are perceived as downtrend are some retracement of a previous medium or long term trend. Since companies are generating revenue, and some of those revenues are being used for the growth of the company, the equity markets are always in a growing rhythm. Since people's psychology enters in the pricing process this growth is not a rational and linear growth. Here the growth model tries to explain this behavior with the help of the terms of medium term trend and long term trend.

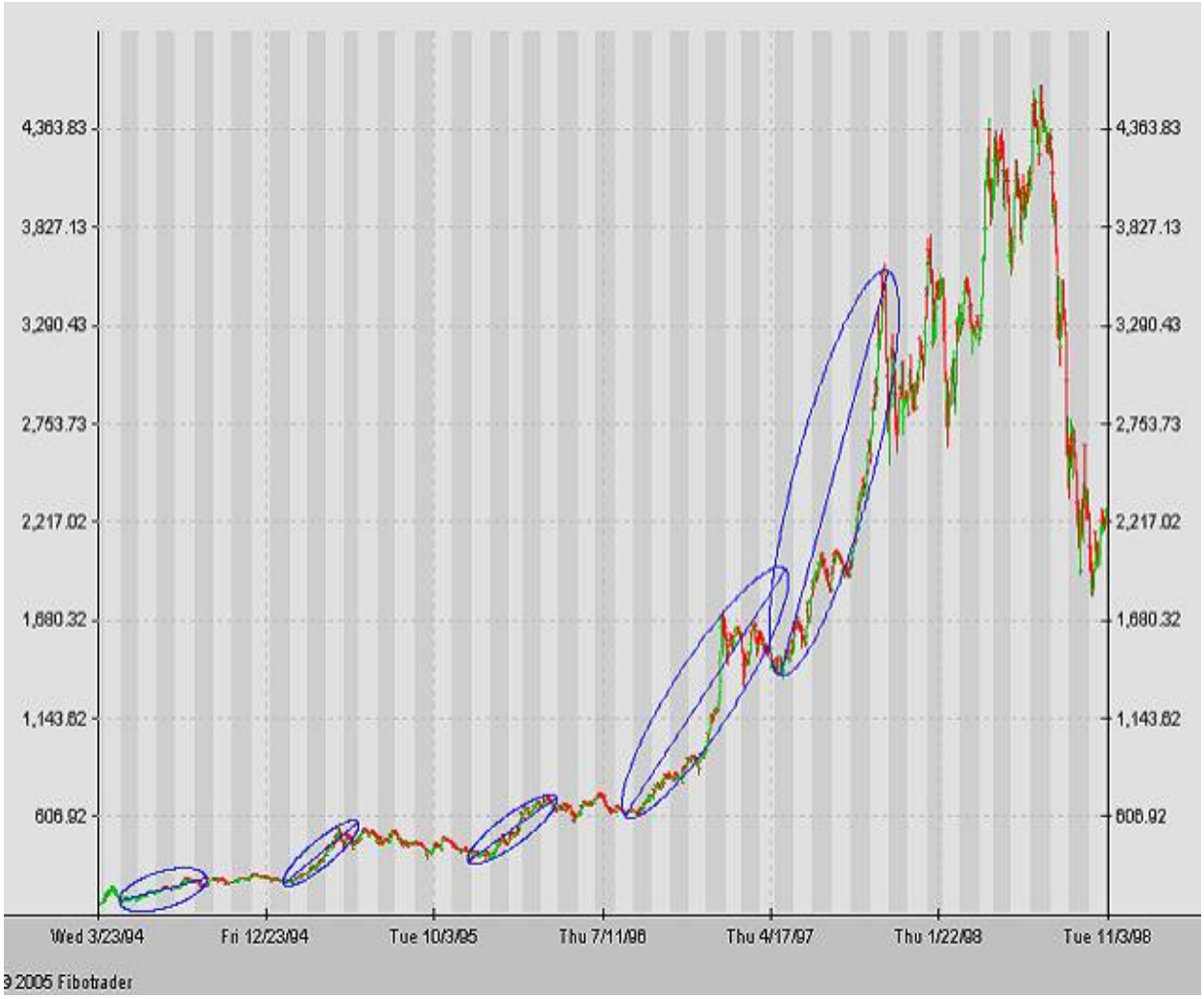
According to the growth model, a long term trend is a situation where more than one medium term trend occurs consecutively. Those medium term trends that form the long term trend are called dependent medium term trends. There are also some times in the market where there is no long term trend and a single medium term trend ends without a consecutive new medium term trend. In that case, those medium term trends are called independent medium term trends. The following examples show those trends in order to understand better.

Figure 1: An Independent Medium Term Trend of ISE100



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Figure 2: A Long Term Trend from ISE100 Including the Dependent Medium Term Trends



The market's growth is formed mostly the contribution of the long term trends and the independent medium term trends. A medium term trend can be divided into three phases. The growing phase, the consolidating phase and the termination phase. The 38.2 % Fibonacci Retracement level is a critical level for a medium term trend. First of all if a medium term trend sees this level, this is the most important evident of the end of the growing phase. The phase which is between the growing phase and the breaking of the Fibonacci 61.8 % fan level is the consolidating phase, after this level has been broken the medium term trend enters to the termination phase.

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Figure 3: The Phases of a Medium Term Trend from ISE100



Generally the dependent medium term trend does not face with a termination phase, because a new medium term trend's growth phase starts on somewhere in the consolidating phase.

Those big moves which are great enough to enter a risk where our trading strategy searches are the medium term trends. Generally the trading strategy tries to enter long positions on the growing phase of the medium term trends and short position on the termination phase of the long term trend.

4 The Trading Strategy

This paper introduces a trading strategy with suitable risk control system with the principal of avoiding risk. Although the principal seems to be concentrated on risk rather than return, the results show that the returns of a few year is much more than the return of the best buy and best sell points in the market, if no leverage is used.

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The first simple principal of the trading strategy is to catch the moves which are big enough to take the risk, which are called as medium term trends in the price behavior.

Technical analysis uses the fact that people behave as a trendy manner, and tries to detect these trends, where the equilibrium of demand and supply is occurring on higher (lower if the trend is downwards) price levels. Casti (2002), states that the human emotions and behaviors are repeated many times in the history, and he adds that;

"The stock market is just one way to take society's emotional temperature. If you look at the average length of hemline as fashions change and plot it against the Dow Jones, there is a striking correlation: the stock market faithfully rises and falls with hemline length. The obvious explanation is that when people are feeling bold and adventurous, they buy stocks and wear more revealing clothes. When they feel threatened and conservative, they do the reverse. The mood is pervasive and almost everyone gets swept along it".

Although the principles of the strategy does not change, some minor adjustments may be needed according to the past price behavior of the market. Generally these adjustments are needed while using the strategy on currency markets instead of equity markets. Understanding the price behavior of the trends in the market is the first step before implementing the strategy. According to our analysis on the Istanbul Stock Exchange and Dow Jones, above behavior of the trends is the reference point while implementing the trading strategy.

5 An Implementation of the Trading Strategy

The strategy will use the Dow Theory's definition of the uptrend in order to catch the medium term trends and enter long position on those trends' growing phase. Since an uptrend is a situation in which, there is a pattern of rising peaks and troughs, the enter signal in the strategy which is defined on this paper, waits for the top of a peak to be broken on a closes basis.

This signal is the first signal of the beginning of the growth phase of a medium term trend, unless the market itself is not in a termination phase of a long term trend. The end of the termination phase of a long term trend is tested with the usage of phi[1] ellipses, which is developed by Fischer (2001).

[1] A Phi Ellipse is a kind of geometric technical tool, in order to define the end of the trends and the turning points in the markets, which is developed by Fischer (2001). The usage seems similar to the linear trend lines but in fact it is much more complex. In a case of irregular trends where the usage of linear trend lines is not suitable, phi ellipses can be used and generate signals for turning points much earlier than the trend lines.

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Figure 4: Break of a peak on the closes basis.



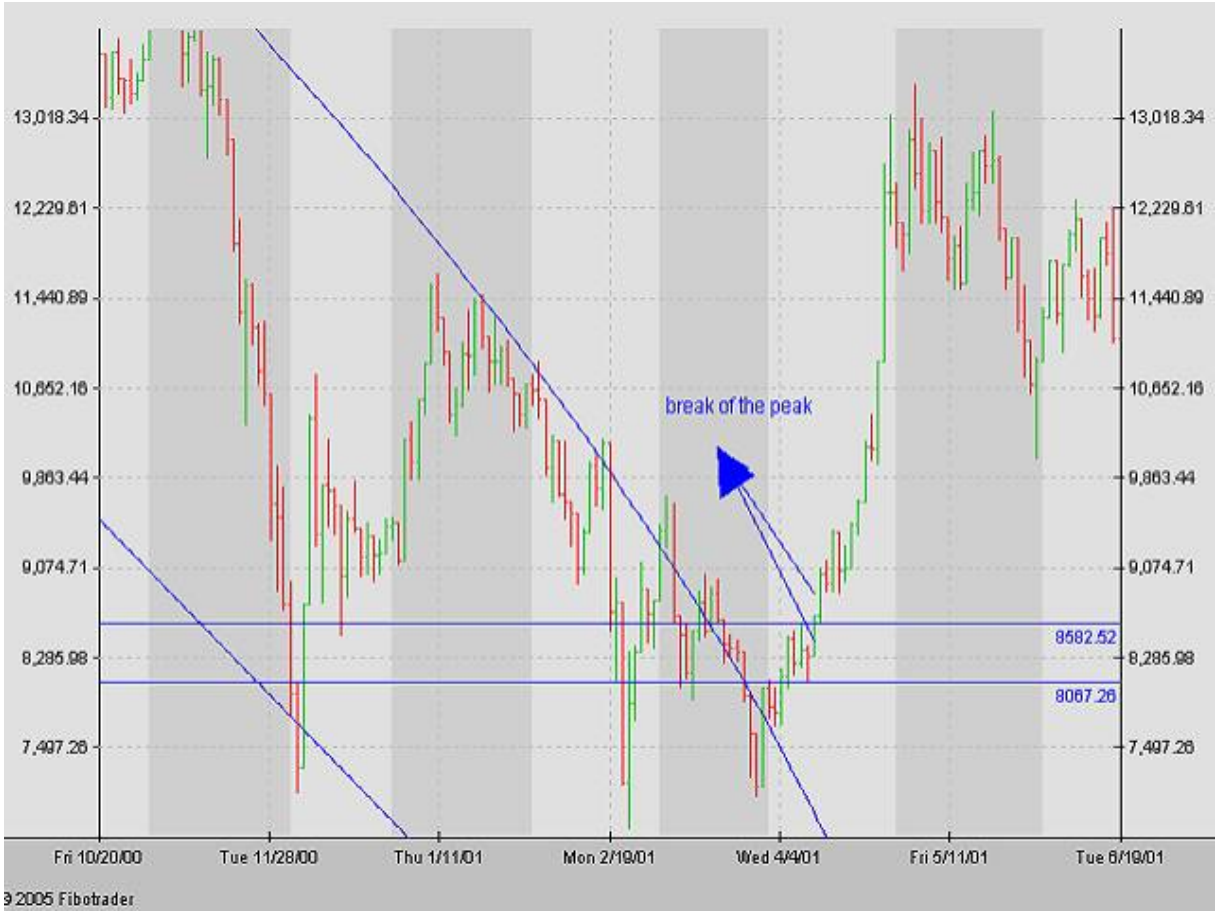
Figure 5: Using the PHI Ellipses in the Termination Phase of a Long Term Trend



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When the phi ellipse is broken on daily closing basis, this is the end of the termination phase of the long term trend. The below figure shows the position taken by a strategy on a long term trend of the market.

Figure 6: Entering long position with the help of phi ellipses.



As mentioned before, the first step in order to take a position is detecting the beginning of a medium term trend. The first signal for the beginning of a medium term trend is the breaking of the top of a peak on the closes basis.

When taking positions, the trading strategy is applied with the following simple risk control method. The main principle of the method is increasing the position when prices are showing that the trade position is right which is generally called pyramid model.

Another important principle is that, the trader should not use all of his equity as margin in a trade. We use here maximum 60 % of equity in trading as margin according to our risk return preferences.

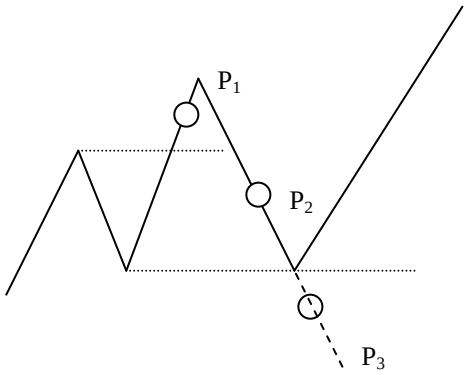
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The third principle comes from the fact that the trader can not be always right on his decisions, but how he will understand that he is wrong. The human psychology leads the trader to close his position too early when losing, although the wrong bet has not been proved by the market yet. That's why the stop loss system should be in harmony with the trading system. The past wrong signals of the trading system should be analyzed with the right signals in order to define the optimum stop loss points.

According to the defined trading strategy on this paper the stop loss point is the breaking of the trough of the first wave which is used also for entering the position.

The behavior of the first wave may not be suitable in order to apply this stop loss efficiently. In a 10 % margin trade the maximum acceptable loss may be 7-8 % change on the market if the appropriate risk control systems are used. The technical stop loss system defined above may not be consistent with the financial stop loss point.

Figure 7: Technical and financial stop losses



P_1: Enter price for long position
P_2: Financial stop loss
P_3: Technical stop loss

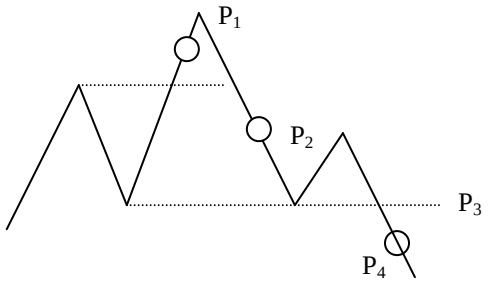
$(P_1 - P_2) / P_1 = 7 – 8$ % which is the maximum acceptable loss
On the other hand, $(P_1 - P_3) / P_1$ may be greater than 10-12 %.

One of the important principles of this paper, claims that the stop loss points should be in harmony with the trading strategy and the trading strategy tells that, the bet of a long position can be accepted as wrong if and only if the trough of the first wave is broken on daily closes basis.

In such a case that the financial stop loss is not consistent with the technical stop loss, this paper suggests the trader to use options in order to apply the technical stop loss.

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Figure 8: Using options on the stop losses



P_4 is the daily closing price that breaks the trough of the first wave which is P_3.

Assume (P_1- P_4) / P_1 = 13 % loss on market meaning that % 130 of the initial margin is lost. (Leverage ratio of 10 is used in the trade) The trader will accept a 7 % market loss on the trade which is (P_1- P_2) / P_1. The market price below P_2 is not acceptable for the trader. In this case, this paper suggests the trader to buy a put option with the strike price of P2 to hedge her position while entering the long position. (P_1 is the spot price while the option is bought, meaning that the option is an out of the money option decreasing the cost of hedging) Since the stop loss level is P4, the position will be closed at P_4 and the trader will not need hedging for below that price level. If the market price comes to P_4, the main position and the option should be closed.

Figure 9: Stop Loss Point on a Long Position



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The implementation of the risk control method on entering long position on the beginning of a medium term trend is below.

Figure 10: Entering Long Position on Medium Term Trend with Risk Control Method



The position is taken on the medium term trend with using the pyramid model. When the first signal is detected a long position is taken with the 10 % of the equity then at the each closing price higher then the previous maximum price, position is increased. In 3 days position is increased to 30 % of the equity. Then it is waited for the prices to break the second peak, and after that position is increased to 60 % of the equity as before. This is the successful example of entering a long position on a medium term trend.

When a long position is taken on a medium term trend, the position is decreased and exited according to usage of phi ellipses on the growth phase of the medium term trend.

This strategy of taking long positions on the growing phase of medium term trends are applied to all medium term trends of a long term trend with the same principles.

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There is also an opportunity of taking a long position on the consolidating phase of a medium term trend. When the past data has been analyzed, it is seen that; after the end of the growth phase generally a strong support level occurs on the 38.2 % Fibonacci levels, in some cases this level also can be 50 % Fibonacci level. After the prices find support on those levels, a rally which goes above the 23.6 % Fibonacci level has been seen. Generally this case makes an opportunity to make 10 % market return meaning that 100 % return on equity. Since the trade is done counter to the market and the time length of the move is not great enough, the pyramid model of risk control method can not be used.

This kind of trade becomes much more risky when compared with the first method of entering long position of the growing phase. That's why according to our risk reward preferences, the trader should not enter this kind of trade with more than 20 % of the equity.
Two kinds of counter market trading method can be suggested according to our analysis on the data. Data shows that market turns mostly on the 38.2 % Fibonacci Retracement level, but sometimes this turning point can be 50 % Fibonacci Retracement level. Since both the data of ISE100 and Dow Jones are market indexes, the volatilities of individual stocks are not seen in the analysis. After the end of the growing phase of a medium term trend, the 50 % Fibo Retracement level is a very strong level for a market return. According to this information the following strategies can be applied for counter market trading.

1) Long position can be taken at the Fibo 38.2 % level and if the market goes below that level, losses are absorbed with the reserve equities with the expectation that the turning point of the market will be minimum 50 % Fibo Retracement level. At least 1.5 times the initial margin should be used as reserve equity in the trading. After the market has turned from that level, it is expected to see a small rally above the 23.6 % Fibo Retracement price level. The exit point in this rally is also defined with the usage of phi ellipses. If the move is not suitable to use a phi ellipse, it is also possible to use linear trend lines for the exit points. This paper suggests the trader to enter such a trade with maximum 20 % of the equity. The weakest side of this method is that there is no stop loss point. In this trade when 20 % of the equity is used as initial margin, 30 % of the equity will be the reserve equity. In a worst case scenario if the market does not turn from the 50 % Fibo Retracement level the trader can lose maximum the reserve equity and 25 % of the initial margin (The amount can be lost before reaching the maintenance margin) which is equal

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to 35 % of the total equity. We did not see this kind of situation in the data, but the maximum risk of the trader is this. Generally market turns and reaches above the 23.6 % Fibo level, which is giving the trader a 10 % market return, which is meaning that 100 % equity return. Since we use the 20 % of the equity as initial margin, the reward of this trading will be 20 % of the total equity. The data shows that, the probability of facing with reward is much more than facing with risk. Figure 11 shows the implementation of this method.

2) Since the above counter market method does not have a stop loss point, it may not be suitable for every trader to bear losses without a stop loss point. That's why another counter market trading method with different enter points and with a stop loss point is defined below.

   First of all it is assumed that the market may return from the 38.2 % Fibo Retracement level. Then support and resistance levels which are above and below the 38.2 % Fibo Retracement levels are defined according to the growing phase of the medium term trend. If the market sees a daily close which is below the first support level but above the second support level and at the following days if the market breaks the first resistance level, this is the enter point for the trade, meaning that 38.2 % Fibo Retracement level has worked as a turning point. After that if prices decrease and if we see a daily closing level which is below the lowest previous price level, this is accepted as a stop loss level. If no enter signal occurs for the 38.2 % Fibo Retracement level and prices approach to the 50 % Fibo Retracement level, than the same principle can also be applied to the 50 % Fibo Retracement level. The Figure 12 shows the implementation of this method to the 50 % Fibo Retracement level.

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Figure 11: Trade Counter to the Market 1

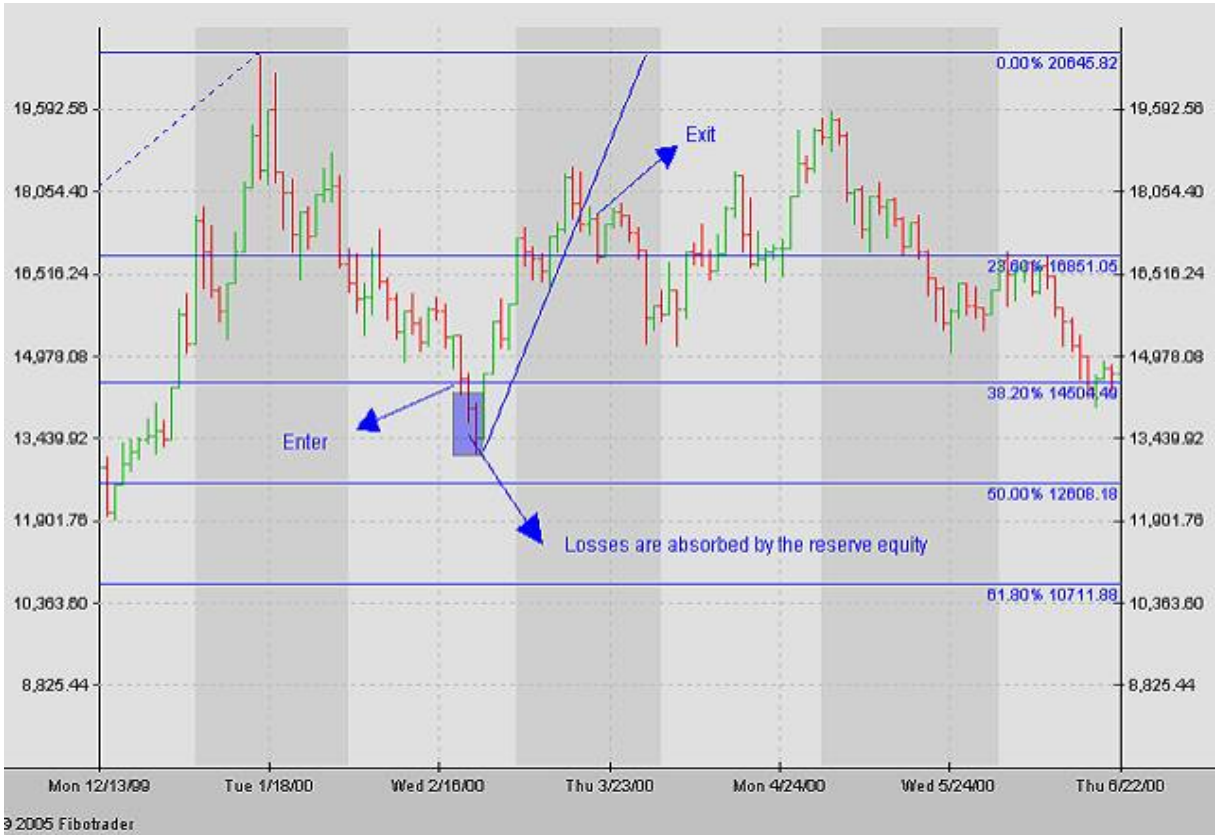
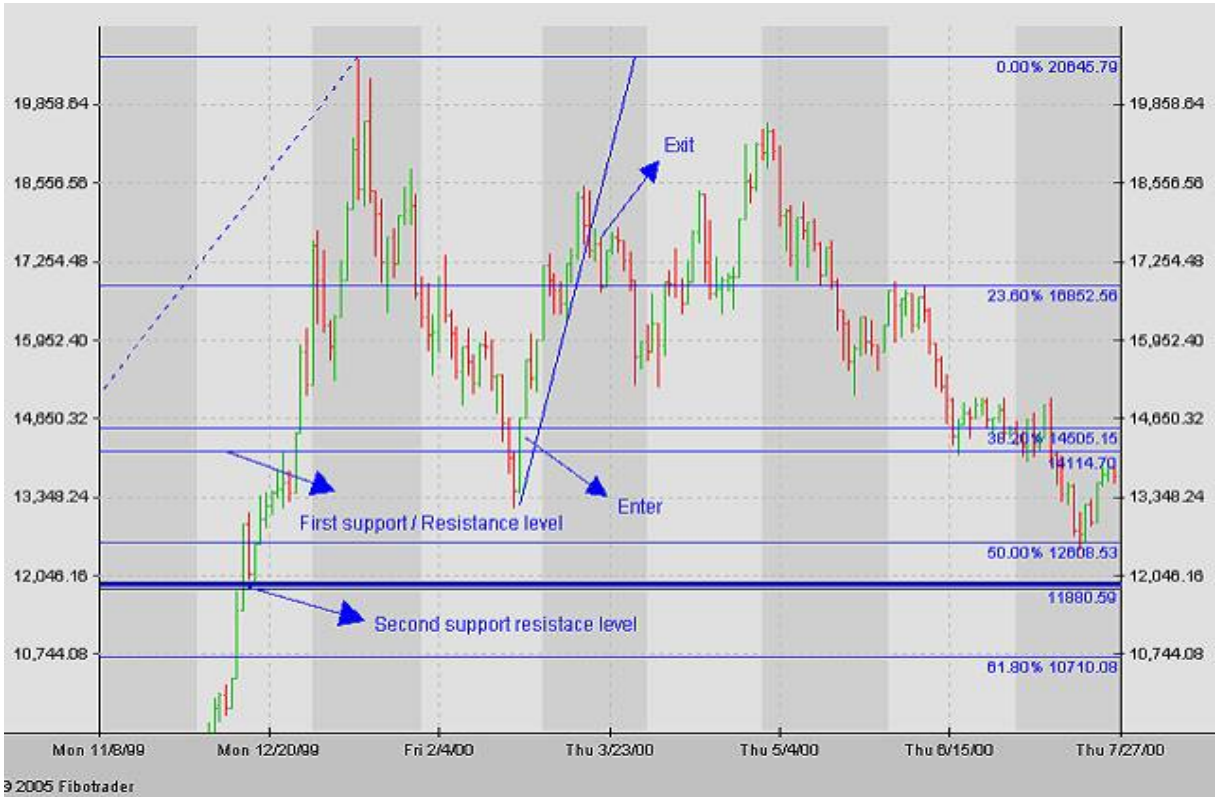


Figure 12: Trade Counter to the Market 2



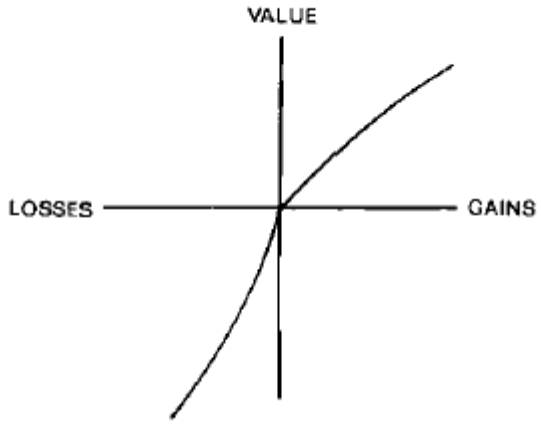
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After the consolidating phase, the signal to enter a next medium term trend is waited. There is also possibility that the first medium term trend that, we have traded before, may not be a part of a long term trend. In that case after the consolidating phase, the medium term trend is expected to enter to the termination phase. The breaking of the 61.8 % Fibonacci Fan level is assumed to be the start of the termination phase. In this case if the trough of the previous move has been broken on closes basis, the trader can enter short position with using the pyramid model more aggressive than the long position.

The analysis on the data shows that the bull markets' upside move is slower than the bear markets' downside move.

Kahneman and Tversky (1979), makes an analysis of decision making under risk and develops an alternative model called prospect theory to Markowitz' expected utility theory. In Prospect Theory, a value function is defined instead of expected utility function, showing the value that people give to the possible gains and losses. The value function is generally concave for gains and convex for losses and it is steeper for losses than for gains. The following figure shows the change in the value function, from losses to gains.

Figure 13: Value Function of Prospect Theory

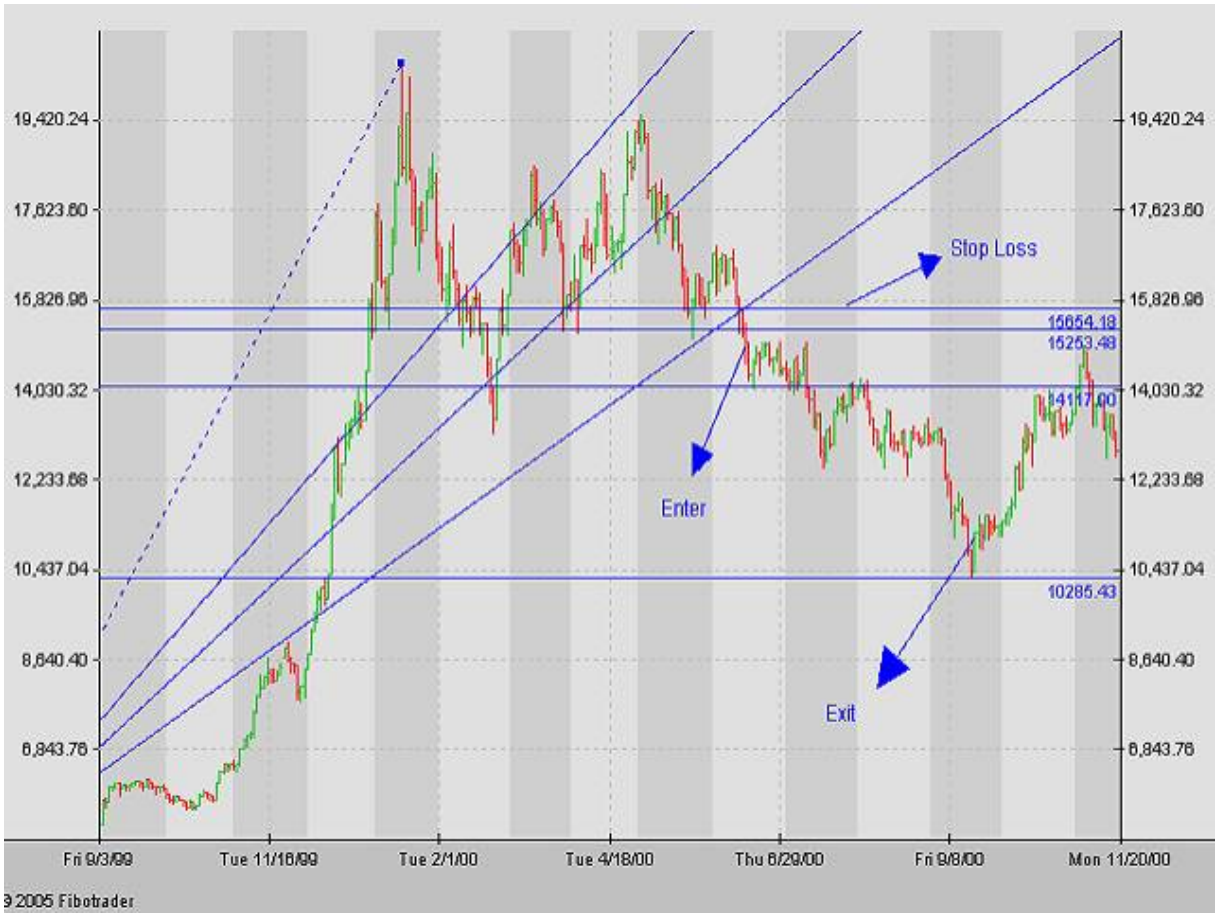


This figure tells that, the same amount of loss gives more pain to human psychology than the same amount of gains' pleasure, showing that people can not bear losses. Every loss gives more pain to the trader and after a psychological barrier is passed people try to get rid of the idea of loss for whatever it worth.

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It is seen that the big percentage of a downside move becomes on small periods of time with panic sales. The above psychology is the reason of this kind of market behavior. When the termination phase of a medium term trend is detected with Fibonacci Fan lines, surprisingly the break of a trough on closes basis gives the signal for those panic sales. Short position should be taken in such a condition. At the first signal this paper suggest to enter the trade with 15 % of the equity, if at the second day the closing price is lower than the first day's closing price then the position should be increased with another 15 % equity. Then the market should be observed and possible support levels should be defined according to the growth phase. If a price reaction occurs on a support level this is the sign of the closing the position. The following figure explains the mentality of the trade better with also showing the stop loss point.

Figure 8: Short Position on Independent Medium Term Trend



Here we do not use the phi ellipses to detect the end of the downside move because of the above behavior. The idea of selling short in this trade is not to enter a long downside move, but to enter the panic sales of very short period and close the position after the panic has been ended. The

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reason of the usage of phi ellipses on long positions is the expectation to enter a trend which is great enough in time and price move.

Generally taking short positions on stock indexes are difficult for traders because the traders try to take short positions on the consolidating phase of the medium term trend. Since in the consolidating phase the market generate volatile moves with no direction, the efforts of taking short positions result with losses and discourage the trader. When the medium term trend enters to the termination phase in which the trader should be ready to enter short position, trader becomes with no power mentally and financially to take a new short position. That's why this method is assumed to be one of the most important aspects of this paper.

A long term trend is defined as the whole move which is formed by minimum two consecutive medium term trends. The critical point on this paper is, to analyze the market as a whole. Before entering a position, the trader should know first, in which part of the long term trend does the market stand. Then he should analyze the medium term trend.

Defined trading strategy on this paper enters the long term trend on the long side with the growth phase of the medium term trends, that's why the trader should not be concerned about entering the long term trends on the long side. On the other hand, the results of the analysis on the data shows that, there is an opportunity of entering short position by concentrating on the termination phase of the long term trend. Generally two kind of termination phase can be seen on the long term trends. The last medium term trend enters to the termination phase and this event triggers the whole long term trend to enter to the termination phase. In such a case the trader will be already on a short position according to the termination phase of the medium term trend. The long term trend which is seen in 2000 on the ISE100 index has this kind of behavior end taking short position according to this method gives successful results. Also most of the long term trend on the Dow Jones index shows this kind of behavior.

Sometimes on the consolidation phase or the termination phase of the last medium term trend, a rapid rally that pushes the prices above the highest point of the last medium term trend can occur. In fact this move is not a medium term trend itself, because the time length of the move will be so small when compared with the medium term trends. Also this move will not be as permanent as the medium term trends. It may collapse so rapidly as it increases, and market can enter a no direction volatile move above the highest point of the last medium term trend. In this

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case it is also possible to use Fibonacci fans on that move (like a medium term trend) and apply the rules of entering short position on the medium term trend.

6 Findings

The above trading strategy is applied to the data of ISE100 and DJI30 indexes between 2000 and 2005 years. In this selected period ISE100 index sees the minimum price of 6850 and maximum price of 28350 as extreme points. A buy and hold investor which is lucky to enter and exit at the extreme points can have a return on equity of 312 %. On the other hand, Dow Jones index sees 7250 points as the minimum level and 11820 as the maximum level, giving the above buy and hold investor maximum 63 % return in 5 years.

The following results are seen on the analysis;
100000 $ on the beginning of 2000, which is traded on DJI30 index according to the above strategy becomes 223500 $ on the end of 2004. The return on equity for four years becomes 123.5%.

On the other hand, 100000 YTL on the beginning of 2000, which is traded on ISE100 index according to the above strategy becomes 141380000 YTL on the end of 2004 with the following assumptions. The return on equity for the following four years is 141.280 %

First of all especially for the second result, it is assumed that, the market is always liquid enough to hold the traded amount, and the amount is small enough in the market for not to manipulate it. The transaction costs are 0.2 % for each contract. The trading prices are done with the closing prices, and it is assumed that there is enough liquidity on the closing prices in order to enter the required positions.

It is obvious that there is a huge difference between the two results on the analysis. The main reasons for this difference are below;

Firstly, the ISE100 index has the advantage of the emergent markets volatility, while the moves on the Dow Jones are limited which is restricting the profit opportunities.

Secondly, before 2000, Dow Jones was more suitable for the principles of growth model and the trading strategy. After the end of the last medium term trend in 1999, Dow Jones has been

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entered a sideway move between 2000 and 2002 causing neither bearish nor bullish market which is suitable for wrong signals. This kind of move is the weakest side of the trading strategy. Some long positions with the expectation to enter a medium term trend, have lasted with loss on the sideway market. On the other hand, in ISE100 there are bullish and bearish markets in 2000 and 2002 giving the opportunity on both long and short positions. After 2004 a long term trend which is the most profitable move for this strategy has been seen on the market. Also normally the behavior of Dow Jones index is more complex than the ISE100 which is making difficult to define the peaks and troughs.

7 Conclusion

Considering the leverage effect in the stocks investments, this study used a growth model for the behavior of the trends in stock markets. A technical trading model and a trading strategy based on Fibonacci approach has been developed by integrating Dow's view for stock markets.

Neither our study in this study nor technical analysis does try to predict the future with the past data of price as some people claim. This study never makes price predictions with technical analysis. Price predictions can be made according to the fundamental scenarios. Despite the widespread idea, there is no need to know where prices may go, in order to take winning positions on the long term. Our study tries to detect the points where probability of opening or closing a successful position is high. Since this is a case of probability there will always be wrong bets but if the appropriate risk control systems are used, in the long term the profitability of the trades will be satisfactory enough. Although the successful results are seen in the analysis as a trading strategy, it may not be suitable for each personality to apply it, because of the need of patience and time to stay on the positions.

Here the discussed trading strategy is not a magical tool in order to beat the stock market but, the methodological presentation of the simple principles of speculation. Two simple but important principle of the speculation is staying on the winning position and closing the losing position. The strategy in this paper applies those principles as efficiently as it can, by using the technical stop loss points to cut the losing position and the usage of phi ellipses in order to stay on the growing phase of the medium term trends while the position is winning.

The results show that, if these principles are applied with discipline in the long term, the leverage effect decreases the dependence of the investors' return on the market return.

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The daily charts are used as an input in the analysis, and the trades are assumed to be done with the daily closing prices, which is decreasing the flexibility of the trader. Especially the usage of the daily prices as an input makes the technical stop loss levels, financially intolerable for the traders if the pyramid risk control method is not used. One criticism for the pyramid risk control method may be its decreasing effect of the capital efficiency. The advantage of the leverage trading is decreased by using this kind of risk control method. In fact the trader does not care the loss in market points. The 2500 points of loss with the size of 1 unit position is the same for the trader 250 points of loss with the size of 10 unit positions. So if the same strategy can be applied with the smaller technical stop loss points, the trader can enter the position with greater sizes but the same value at risk. The one way to generate smaller technical stop loss points is using the 60 min and 5 min charts in harmony with the daily charts. This study can be developed with further research on this topic.

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